

F78PB Probability and Statistics B

Hypothesis Testing

Two samples: Differences between means

Test H_0: $\mu_1 = \mu_2$
versus H_1: $\mu_1 \neq \mu_2$ (or $\mu_1 > \mu_2$, etc.)
at a specified significance level α.

Assuming that the samples are normal and independent we can use the following test statistic

$$Z = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}} \quad \sim \mathrm{N}(0, 1)$$

with s_1^2, s_2^2 substituting σ_1^2, σ_2^2 if unknown, providing both n_1 and n_2 are large enough.

With small n_1, n_2,we need to assume a common unknown variance, σ^2 for both populations, estimated by

$$s_p^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}$$

and use the statistic

$$T = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{S_p\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} \sim t_{n_1+n_2-2}$$

We again need to assume that both populations are normal.

Two dependent samples

When the samples are not independent and paired, i.e.

$(x_1, y_1), (x_2, y_2), \ldots\ldots, (x_n, y_n)$, we can work with the differences $d_i = x_i - y_i$ and the mean $\mu = \mu_x - \mu_y$.

Test H_0: $\mu = 0$

versus H_1: $\mu \neq 0$ (or $\mu > 0$, etc.)

using the statistics presented before for the single sample situation,

e.g. $t = \frac{\bar{d}}{s/\sqrt{n}} \sim t_{n-1}$ for small n.

($\bar{d}$ is the sample mean of the d_i values and s is their standard deviation). Here, we need to assume that the differences come from a normal population.

Hypothesis Testing For Variances

Single sample situation

Test H_0: $\sigma^2 = \sigma_0^2$

versus H_1: $\sigma^2 \neq \sigma_0^2$ (or $\sigma^2 > {\sigma_0}^2$, etc.)

at a specified significance level α.

We use the test statistic

$$\chi^2 = \frac{(n-1)S^2}{{\sigma_0}^2} \sim \chi^2_{n-1}$$

Two samples (equality of two population variances)

Test H_0: $\sigma_1^2 = \sigma_2^2$ (or $\frac{\sigma_1^2}{\sigma_2^2} = 1$)

versus H_1: $\sigma_1^2 \neq \sigma_2^2$ (or $\sigma_1^2 > {\sigma_2}^2$, etc.)

The random samples are assumed **normal** and **independent**.

We use the test statistic

$F = \frac{S_1^2}{S_2^2} \sim F_{n_1-1,n_2-1}$

Example

The following two samples have been collected from normal populations with unknown parameters.

$$x = (28, 26, 18, 16)$$
$$y = (30, 32, 26, 24)$$

Test the hypothesis that the means of the two populations are equal.

Hypothesis Testing For Proportions

Hypothesis concerning the proportion p of a population having a specific characteristic.

Test $H_0: p = p_0$

versus $H_1: p \neq p_0$ (or $p > p_0$, etc.)

With large sample size n, use the test statistic

$$Z = \frac{X - np_0}{\sqrt{np_0(1-p_0)}} \sim \mathrm{N}(0,1)$$

Where X is the 'number of successes in n trials' in the sample. Continuity correction may also be used.

For small n we can use the cumulative distribution function of the binomial distribution to obtain the p-value.

Example (Belgian 1 Euro Coin)

The Guardian reported on January 4th 2002 that two Polish mathematicians claim that the Belgian one euro coin is ‘struck asymmetrically’. They spin the Euro coin 250 times and found it landed heads up 140 times.

Is their claim statistically justified?

Two-Sample Test for Proportions (Large Samples)

For two independent samples with large n_1, n_2, use the test statistic:

$$Z = \frac{\frac{X_1}{n_1} - \frac{X_2}{n_2}}{\sqrt{\hat{P}(1-\hat{P})\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}} \sim \mathcal{N}(0,1),$$

where the common proportion estimate is:

$$\hat{P} = \frac{X_1 + X_2}{n_1 + n_2}.$$

Example

In a psychological experiment, a collection of single people were sent £500 but the address on the letter was that of their neighbour.

- Out of 69 male recipients, 17 of them kept the £500.
- Out of 131 female recipients, 11 of them kept the £500.

Is there any difference in the behaviour of the two genders?

Let X_1 denote the number of male recipients who keep the money, and X_2 denote the number of female recipients who keep the money.

Solution

We test

$$H_0 : p_1 = p_2 \text{ versus } H_1 : p_1 \neq p_2$$

using the test statistic

$$Z(X_1, X_2) = \frac{\frac{X_1}{n_1} - \frac{X_2}{n_2}}{\sqrt{\hat{P}(1-\hat{P})\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}, \quad \hat{P} = \frac{X_1 + X_2}{n_1 + n_2}.$$

Our data is

$$n_1 = 69, x_1 = 17, n_2 = 131, x_2 = 11.$$

So our estimate of a common proportion, $\hat{p}$, is

$$\hat{p} = \frac{x_1 + x_2}{n_1 + n_2} = \frac{17 + 11}{69 + 131} = 0.14$$

and hence our observed value of the test statistic is

$$Z(x_1, x_2) = \frac{\frac{17}{69} - \frac{11}{131}}{\sqrt{0.14(1-0.14)\left(\frac{1}{69} + \frac{1}{131}\right)}} = 3.15$$

We test

$$H_0 : p_1 = p_2 \text{ versus } H_1 : p_1 \neq p_2$$

using the test statistic

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Our observed value of the test statistic is

$$Z(x_1, x_2) = \frac{\frac{17}{69} - \frac{11}{131}}{\sqrt{0.14(1-0.14)\left(\frac{1}{69} + \frac{1}{131}\right)}} = 3.15$$

We can use R to find the p-value for this

$$\mathbb{P}(Z \geq 3.15 \text{ or } Z \leq -3.15) = 2\mathbb{P}(Z \geq 3.15) = 0.00165.$$

Therefore we have very strong evidence to reject the null hypothesis.